# Отчёт по практической работе №2

1. На каждом интерфейсе маршрутизаторов задать ip-address из сетей на /30!!!

Интерфейсы роутера C-R-3-1:

```
interface Ethernet0/0
 description "DMikheeva-C-R-3"
 ip address 16.80.1.1 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-VPC-50"
 ip address 169.31.1.2 255.255.255.0
!
```

Интерфейсы роутера C-R-1-3:

```
interface Ethernet0/0
 description "DMikheeva-C-R-1"
 ip address 17.38.3.1 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-VPC-53"
 ip address 172.14.3.2 255.255.255.0
!
```

Интерфейсы роутера C-R-4-1:

```
interface Ethernet0/0
 description "DMikheeva-C-R-4"
 ip address 10.166.10.1 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-VPC-64"
 ip address 10.187.11.2 255.255.255.252
!
```

Интерфейсы роутера C-R-2-3:

```
interface Ethernet0/0
 description "DMikheeva-C-R-2"
 ip address 18.122.3.1 255.255.255.252
!
interface Ethernet0/1
 description "DMikheeva-VPC-57"
 ip address 182.92.3.2 255.255.255.0
!
```

2. На каждом роутере прописать статические маршруты так, чтобы каждое сетевое устройство (Роутеры, VPC, Linux-сервера и коммутаторы), в рамках 1й и 2й лабораторной работы, могло пропинговать друг друга.

Трассировка от C-R-3-1 до C-R-2-3:

```
DMikheeva-C-R-3-1#traceroute 18.122.3.1
Type escape sequence to abort.
Tracing the route to 18.122.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 16.80.1.2 1 msec 3 msec 2 msec
  2 10.38.0.2 8 msec 2 msec 2 msec
  3 10.101.0.1 2 msec 6 msec 2 msec
  4 18.122.3.1 5 msec 3 msec *
```

Пинг коммутатора Switch-3 с роутера C-R-3-2:

```
DMikheeva-C-R-3-2#ping 10.210.70.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.210.70.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

Трассировка от C-R-3-3 до C-R-1-1:

```
DMikheeva-C-R-3-3#traceroute 17.38.1.1
Type escape sequence to abort.
Tracing the route to 17.38.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 16.80.3.2 2 msec 2 msec 1 msec
  2 10.38.0.2 1 msec 2 msec 1 msec
  3 10.59.0.1 2 msec 2 msec 4 msec
  4 17.38.1.1 5 msec 5 msec *
```

Пинг компьютера PC34 с роутера C-R-1-2:

```
DMikheeva-C-R-1-2#ping 10.210.20.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.210.20.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/4/9 ms
```

Трассировка от C-R-4-1 до C-R-1-3:

```
DMikheeva-C-R-4-1#traceroute 17.38.3.1
Type escape sequence to abort.
Tracing the route to 17.38.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.166.10.2 1 msec 2 msec 1 msec
  2 10.122.0.2 2 msec 2 msec 1 msec
  3 10.59.0.1 2 msec 2 msec 2 msec
  4 17.38.3.1 2 msec 2 msec *
```

Пинг SSH-Client с роутера C-R-4-2:

```
DMikheeva-C-R-4-2#ping 10.210.70.200
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.210.70.200, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/4/7 ms
```

Трассировка от C-R-4-3 до C-R-2-1:

```
DMikheeva-C-R-4-3#traceroute 18.122.1.1
Type escape sequence to abort.
Tracing the route to 18.122.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 30.78.3.2 4 msec 2 msec 2 msec
  2 10.122.0.2 2 msec 6 msec 3 msec
  3 10.101.0.1 6 msec 3 msec 3 msec
  4 18.122.1.1 2 msec 2 msec *
```

Пинг компьютера PC16 с роутера C-R-2-2:

```
DMikheeva-C-R-2-2#ping 10.210.50.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.210.50.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/7 ms
```

3. Настроить сеть на VPC с заданием статического IP и Default GW и сохранить настройки.

IP-адрес на VPC50:

```
VPC50> sh ip

NAME        : VPC50[1]
IP/MASK     : 169.31.1.1/24
GATEWAY     : 169.31.1.2
DNS         :
MAC         : 00:50:79:66:68:47
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

IP-адрес на VPC53:

```
VPC53> sh ip

NAME        : VPC53[1]
IP/MASK     : 172.14.3.1/24
GATEWAY     : 172.14.3.2
DNS         :
MAC         : 00:50:79:66:68:4a
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

IP-адрес на VPC64:

```
VPC64> sh ip

NAME        : VPC64[1]
IP/MASK     : 10.187.11.1/30
GATEWAY     : 10.187.11.2
DNS         :
MAC         : 00:50:79:66:68:53
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

IP-адрес на VPC57:

```
VPC57> sh ip

NAME        : VPC57[1]
IP/MASK     : 182.92.3.1/24
GATEWAY     : 182.92.3.2
DNS         :
MAC         : 00:50:79:66:68:4e
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

4. Настроить сеть на Linux серверах с заданием статического IP и Default GW.

IP-адрес на OpenVPN-L2:

```
root@localhost:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:00:00:56:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 30.78.22.1/24 brd 30.78.22.255 scope global ens3
       valid_lft forever preferred_lft forever
    inet6 fe80::250:ff:fe00:5600/64 scope link
       valid_lft forever preferred_lft forever
3: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:00:00:56:01 brd ff:ff:ff:ff:ff:ff
    altname enp0s4
    inet 30.78.11.2/24 brd 30.78.11.255 scope global ens4
       valid_lft forever preferred_lft forever
    inet6 fe80::250:ff:fe00:5601/64 scope link
       valid_lft forever preferred_lft forever
root@localhost:~#
```

IP-адрес на OpenVPN-L3:

```
root@localhost:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:00:00:3a:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 169.31.2.1/24 brd 169.31.2.255 scope global ens3
       valid_lft forever preferred_lft forever
    inet6 fe80::250:ff:fe00:3a00/64 scope link
       valid_lft forever preferred_lft forever
3: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:50:00:00:3a:01 brd ff:ff:ff:ff:ff:ff
    altname enp0s4
    inet 30.78.1.2/30 brd 30.78.1.3 scope global ens4
       valid_lft forever preferred_lft forever
    inet6 fe80::250:ff:fe00:3a01/64 scope link
       valid_lft forever preferred_lft forever
root@localhost:~#
```